

Explanations Leak Less Than Assumed: A Channel Decomposition of Explanation-Guided Model Extraction

Anonymous Author(s)

Abstract

Machine-learning services increasingly return a post-hoc *explanation* alongside each prediction, in the name of transparency and regulatory compliance. We ask how much this practice helps an adversary *steal* the model. Studying explanation-guided model extraction against interpretable classifiers, we decompose what an explanation can contribute to an attacker’s query synthesis into two channels: a feature-*attribution* channel (which features matter) and a decision-*threshold* channel (where along a feature the prediction changes). Through controlled ablations, each evaluated over ten independent target instantiations, we find that the *attribution* channel—the genuinely model-specific part of an explanation—confers no systematic advantage for either explainer: choosing features by SHAP importance is no better than random, and LIME’s feature ranking is centred on zero and inconsistent in sign across targets. The one channel that reliably helps is the decision *threshold* that LIME’s discretization exposes (a bin edge), worth up to 41 fidelity points and, unlike attribution, directionally consistent. That threshold, however, is a quantile of the background data rather than a property of the model: refitting the discretizer on data the adversary already holds reproduces the gain in full, leaving the advantage of the target’s own discretizer statistically indistinguishable from zero. Consistently, an adversary given *no* explanation content—random feature choice and a self-computed grid—matches or exceeds the full-explanation baseline attack. What an explanation contributes to extraction is therefore query-placement convenience rather than disclosure. We do not claim explanations are harmless: our results cover one attack family, tabular data, and two explainers in their default configurations. Within that scope, however, the privacy cost of publishing an explanation is considerably smaller than the literature assumes.

Keywords

model extraction, model stealing, explainable AI, SHAP, LIME, machine-learning-as-a-service, privacy

1 Introduction

Machine-learning services increasingly return a post-hoc *explanation* alongside each prediction. Cloud providers ship attribution APIs as standard tooling, and regulatory pressure—the GDPR’s provisions on automated decision-making, and more recently the EU AI Act’s transparency obligations—pushes deployments further in that direction. The premise is that a prediction accompanied by a reason is more accountable than a prediction alone.

That premise has a security cost, because an explanation is additional output computed from the very model an adversary would like to steal. Model extraction reconstructs a functional copy of a black-box target from query access [4, 16], and the resulting surrogate is not merely an intellectual-property loss: it is a platform for further attacks, since a functionally equivalent copy enables membership inference at a fraction of the direct query cost [3, 11]. If explanations make extraction materially cheaper, then transparency and privacy are in direct tension, and providers face a genuine trade-off.

Explanation-guided extraction attacks report exactly that: querying with explanations attached yields higher-fidelity surrogates for fewer queries [1, 8, 10]. These results establish *that* explanations help, and are typically reported end-to-end: the attack is run with the explanation and compared against a baseline without it. The gain is then naturally credited to the explanation’s model-specific content—which features the model relies on locally, or, for gradient explanations, the local geometry of the decision boundary. An end-to-end comparison, however, cannot separate that content from the other things an explanation supplies, such as a convenient set of coordinates at which to place the next query. Which of these is responsible matters practically, because it determines whether the risk attaches to explanations as such, to particular explanation methods, or to something an adversary could obtain anyway.

This paper. We decompose what an explanation can contribute to an attacker’s query synthesis into two channels: an *attribution* channel (*which* features are locally influential) and a *threshold* channel (*where* along a feature the prediction changes). The distinction is not merely conceptual—it separates the two dominant explanation families. SHAP reports signed magnitudes and exposes only the first; LIME, which discretizes continuous features before fitting its local surrogate, exposes both, since its bin edges are candidate decision thresholds. We ablate each channel independently inside a state-of-the-art explanation-guided attack, holding the traversal, the seeds, and the random state fixed, and evaluate every comparison over ten independent target instantiations.

Our findings are largely negative, and we think that is the useful result. The attribution channel—the part of an explanation that is genuinely model-specific—confers no systematic advantage for either explainer: replacing the explanation’s feature ranking with a random choice leaves fidelity centred on zero, with significant effects in *both* directions depending on the target. The one channel that reliably helps is LIME’s discretization threshold, which is worth up to 41 fidelity points. But that threshold is a quantile of the background data, not a property of the model: when we refit the discretizer on data the adversary already holds, the gain is reproduced in full, and the advantage of the target’s own discretizer is statistically indistinguishable from zero. Taking both channels together, an adversary that reads *nothing* from the explanation—selecting features at random and snapping to a grid estimated from its own

This work is licensed under the Creative Commons Attribution 4.0 International License. To view a copy of this license visit <https://creativecommons.org/licenses/by/4.0/> or send a letter to Creative Commons, PO Box 1866, Mountain View, CA 94042, USA.

Proceedings on Privacy Enhancing Technologies 2027(X), 1–15

© 2027 Copyright held by the owner/author(s).

<https://doi.org/XXXXXXX.XXXXXXX>



auxiliary sample—matches, and on several targets exceeds, the full-explanation baseline attack. What an explanation contributes to extraction is therefore *query-placement convenience*, not disclosure.

Contributions.

- A two-channel decomposition of explanation leakage (attribution versus threshold) that makes “what does an explanation leak” an answerable empirical question, and a protocol for testing it that isolates each channel while holding the rest of the attack fixed (§3, §4).
- Evidence that the attribution channel confers no systematic extraction advantage—for SHAP and for LIME alike—and that where explanation-guided selection does have a significant effect, its *sign* is a property of the target rather than of the explanation (§6.4).
- Evidence that the threshold channel, which does help substantially, is *self-computable*: an adversary holding a modest auxiliary sample reconstructs an equivalent discretization with no measurable loss, so the channel is not a disclosure by the service (§6.5, §6.6).
- A boundary-and-diversity query-synthesis layer used as the measurement instrument, ensuring that null results are not artefacts of a weak adversary (§4, §6.2).

Scope. Our conclusions are bounded in ways worth stating up front. We study one family of explanation-guided attack—traversal-based extraction against interpretable classifiers [10]—on tabular data, with two explainers in their default configurations. A differently designed attack, or a discretizer that depends on labels rather than on the marginal distribution, could plausibly extract more from the same explanations. We therefore claim that post-hoc explanations leak considerably less than the literature assumes *in this setting*, not that they are harmless in general.

2 Background and Related Work

2.1 Model Extraction and Post-hoc Explanations

Model extraction attacks reconstruct a functional copy $\hat{f}$ of a black-box target f from query access alone, with success measured by *prediction agreement* between $\hat{f}$ and f on held-out data (formalised in §3). When the target is deployed as an explainable service, each query returns not only a label but a post-hoc *explanation*, which may reveal additional structure about f . Two dominant explanation families expose fundamentally different information:

- **Attribution methods (SHAP).** SHAP [7] assigns each feature a signed magnitude $\phi_i(x)$ —its Shapley marginal contribution to the prediction at x . This is a statement of *how much* each feature matters locally; it carries no explicit information about *where* along a feature axis the decision changes.
- **Local surrogate methods (LIME).** LIME fits an interpretable model to a *discretized* neighbourhood of x : continuous features are binned, and the returned coefficients are attached to (feature, bin) pairs. LIME therefore reveals both an influential feature *and* the bin edge—a *threshold*—that separates neighbouring regions.

This distinction between *magnitude-only* and *threshold-bearing* explanations is central to our analysis: as we show, it is the threshold, not the attribution magnitude, that an extraction attacker can exploit (§3, §5).

2.2 Related Work

Model Extraction Attacks. Model extraction attacks aim to reconstruct a functional copy of a black-box target model through strategic querying [16]. Early work by Tramèr et al. [16] demonstrated that simple equation-solving over prediction outputs suffices for linear models. Subsequent methods scaled to neural networks: Knockoff Nets [12] trains a surrogate on outputs obtained by querying with a knock-off dataset, while Jagielski et al. [4] show that high-fidelity extraction of deep networks is possible given sufficient queries. A parallel line of work [2, 15] frames model extraction as knowledge distillation, transferring decision boundaries through synthetic queries. Marich [5] formalises the problem as *distributional equivalence* and proposes an active sampling strategy that maximises information gain per query. In contrast, our method requires only a small per-class seed set rather than a large public transfer set, and synthesises informative queries from that seed.

Explanation-Guided Extraction. The observation that post-hoc explanations leak structural information about the target model motivates a distinct attack family. Milli et al. [8] first demonstrated that gradient-based explanations suffice to reconstruct linear models exactly. Autolytus [10] extended this to classical ML classifiers (decision trees, logistic regression, naive Bayes, and k -NN) by using LIME and SHAP explanations as traversal guides: the attacker perturbs seed samples along the directions of highest feature attribution, progressively labelling a synthetic dataset to train a surrogate. Yan et al. [18] (XaMEA) similarly exploit the spatial information contained in explanations in a model-agnostic manner. A common assumption, implicit in these attacks, is that the explanation’s *attribution ranking* is what drives the extraction gain. Through controlled ablations we revisit this assumption and find it does *not* hold: the gain stems primarily from boundary-directed query synthesis—boundary search and diverse sampling—while the choice of features by attribution contributes little beyond a random choice. The explanation supplies additional signal only when it exposes a decision *threshold*: LIME’s discretization surfaces such thresholds directly, whereas SHAP’s magnitude attribution does not, so a SHAP-guided attacker must *search* for the boundary and gains little over the sampling procedure itself. This reframes explanation leakage as a matter of decision *thresholds* rather than boundary *geometry*.

Counterfactual-Based Extraction. A growing body of work exploits counterfactual explanations (CFs) rather than attribution scores. Aivodji et al. [1] first showed that CFs leak enough boundary information to build high-fidelity surrogates, and DualCF [17] improved query efficiency by training on CF/counter-CF *pairs* that straddle the decision boundary. Khouna et al. [6] propose the Tree Reconstruction Attack (TRA), which uses a *restricted* CF oracle to perform axis-aligned binary search, achieving exact functional reconstruction of decision trees and random forests. Although theoretically elegant, TRA requires an oracle that returns the *nearest*

counterfactual confined to a specific hyper-rectangle—an assumption rarely satisfied by deployed XAI APIs. Concurrently, Otto et al. [13] derive query complexity bounds for extracting linear classifiers under factual and counterfactual query regimes. Our work connects these lines: rather than a bespoke CF oracle, we obtain boundary thresholds from the explanation itself—directly from LIME’s discretization, or by SHAP-directed single-axis bisection—without requiring restricted oracle access. Consistent with our central finding, it is this threshold signal, not the attribution magnitude, that constitutes the exploitable channel.

Privacy Implications of Explanations. Recent work highlights the security cost of XAI transparency. Ezzeddine et al. [3] show that counterfactual explanations amplify membership inference attacks (MIAs) in MLaaS settings, characterising a fundamental utility–privacy trade-off. LoMime [11]—from the authors of Autolytus—uses surrogate model extraction as a stepping stone for label-only MIA, demonstrating that functionally equivalent surrogates enable downstream privacy attacks with far fewer queries than direct methods; our extraction fidelity thus serves as an upstream component for such attacks. Surveys of privacy-preserving explanation [9] catalogue countermeasures. Localising the effect to a specific channel (the discretization threshold) lets us ask what such a countermeasure would have to accomplish; we find that obfuscating that channel is unlikely to suffice, since the adversary can reconstruct an equivalent grid from its own data (§6.6).

3 Problem Formulation

Setting. Let $f : \mathcal{X} \rightarrow \mathcal{Y}$ be a black-box target classifier, with $\mathcal{X} \subseteq \mathbb{R}^d$ and $\mathcal{Y} = \{0, \dots, C-1\}$. The attacker knows neither f ’s architecture nor its training data, and interacts with f only through an *explanation-augmented prediction API*. Each query $x \in \mathcal{X}$ returns (i) the predicted label $f(x)$ and the class-probability vector $\hat{p}(x) \in \Delta^C$, and (ii) a post-hoc explanation $e(x)$ produced by a fixed explainer $\mathcal{E} \in \{\text{SHAP}, \text{LIME}\}$. Every call counts against a total query budget Q . The attacker additionally holds a small per-class *seed set* $\mathcal{D}_0 = \{(x_i, y_i)\}$ with n examples per class (following the respective Autolytus conventions, $n=1$ for LIME and $n=5$ for SHAP), drawn from a public split that shares $\mathcal{X}$ ’s feature schema; this is a handful of seeds, not a large public transfer set.

Explanation oracle. The two explainers expose structurally different information about f near x :

- **SHAP:** $e(x) = \phi(x) \in \mathbb{R}^d$, a vector of signed Shapley attributions in which $|\phi_i(x)|$ measures the local influence *magnitude* of feature i [7]. It reports *how much* each feature matters, but not *where* the decision changes along it.
- **LIME:** $e(x) = \{(i, [l_i, u_i], w_i)\}_{i \in S}$ over a selected feature subset S , where feature i has local weight w_i and lies in the discretization bin $[l_i, u_i]$. Beyond a magnitude, LIME therefore discloses a *bin edge* which is an estimate of a coordinate at which f ’s output changes.

Objective. The attacker trains a surrogate $\hat{f} : \mathcal{X} \rightarrow \mathcal{Y}$ that is *functionally equivalent* to f , measured by prediction agreement:

$$\text{Sim}(\hat{f}, f) = \mathbb{E}_{x \sim P_{\mathcal{X}}} [\mathbf{1}(\hat{f}(x) = f(x))]. \quad (1)$$

Here $P_{\mathcal{X}}$ is the (unknown) data distribution over $\mathcal{X}$, estimated in practice by a held-out test set disjoint from the attacker’s seed set $\mathcal{D}_0$ and its queries. The attack succeeds if $\text{Sim}(\hat{f}, f)$ is high while the number of API calls stays within Q .

What does an explanation leak? The power of an extraction attack rests on *query placement*: labelled queries concentrated at the decision boundary and spread across diverse decision regions are far more informative than interior or random queries, independent of any explanation [14]. Our question is what a post-hoc explanation *adds* to this placement. We decompose the explanation’s possible contribution into two channels:

- (1) **Attribution:** *which* features are locally influential (a ranking by $|\phi_i|$ or $|w_i|$), available from both SHAP and LIME.
- (2) **Threshold:** *where* along a feature axis the prediction changes, i.e. the coordinate of the decision boundary. LIME’s bin edge $\{l_i, u_i\}$ is a direct estimate; SHAP’s magnitude carries no such location.

[TODO: we may need to change this paragraph depending on the randfeat + aux results. we can make it stronger if the results are good] Neither channel is an attack in itself. The query-placement machinery of §4—boundary search and diverse generation—is a standard extraction technique that the explanation does not disclose and that an attacker can run without any explanation at all. What an explanation can change is only *where* that machinery directs its queries: the attribution channel proposes which coordinate to move along, and the threshold channel proposes how far to move it. Isolating these two inputs is therefore the natural way to ask what the explanation itself contributes. Prior explanation-guided attacks [8, 10] implicitly credit their gain to the *attribution* channel and, for gradient explanations, to recovering boundary *geometry* (the local normal). Our central hypotheses, examined in §5 and §6, are:

H1 (attribution is inert). Steering query placement by explanation importance does not improve fidelity over a random feature choice. The attack’s gain comes from query placement itself, not from the attribution channel.

H2 (the threshold is the operative channel). The only explanation-supplied signal that improves fidelity is the threshold channel. A threshold-bearing explainer (LIME) helps, whereas a magnitude-only explainer (SHAP) does not, because—lacking a threshold—it must *search* for the boundary.

H3 (the threshold is not a disclosure). The threshold that helps is a statistic of the *data distribution*—LIME’s default bin edges are quantiles of its background sample, independent of the model and of the labels—rather than a property of the target. An adversary holding a modest auxiliary sample can therefore reconstruct an equivalent grid, so the channel accelerates extraction without the service disclosing anything the adversary could not compute itself.

H1 and H2 together locate *which* part of an explanation an attacker consumes; H3 asks whether consuming it required the explanation at all. That last question is the organising principle of this paper: leakage should be measured against what the adversary could obtain unaided, not against nothing.

Threat model. We assume: (1) black-box access only (no white-box gradients); (2) the API returns post-hoc explanations (SHAP or LIME), as increasingly expected for transparency (e.g. GDPR Article 22); (3) the attacker knows the feature schema but not the model family or hyperparameters; and (4) the attacker holds a small auxiliary sample D_A drawn from the target’s distribution, from which it draws $n \leq 5$ seeds per class. Assumption (4) is inherited from the attack we analyse [10], which allows D_A to range “from a rudimentary collection of samples to comprehensive datasets”; it matters here because H3 turns on what the adversary can estimate from D_A alone. This captures realistic MLaaS deployments in which a vendor exposes explanations for regulatory compliance without revealing the underlying model.

4 Method

[TODO: [AC] Demote this section: it is the measurement instrument, not a contribution. Pending the crop blinding result, reframe Phases 1–2 as query-placement machinery whose effect does not depend on explanation input (supported if ours $\approx$ ours_{auxbin}), and drop “diverse generation” from any headline framing – E2/E3/E5 all run on base Autolycus, which has no Phase 1 at all.]

Autolycus expands a small per-class seed set by local $\pm \epsilon$ perturbation, which leaves two systematic gaps that we target directly. First, the surrogate and target disagree most sharply *at the decision boundary*: concentrating labelled queries there gives the surrogate a much sharper estimate of the target’s decision surface than the same budget spent in class interiors. Second—and less obvious—we observe that many surrogate–target disagreements occur on inputs the target classifies with *high* confidence, i.e. confident interior errors rather than boundary ambiguity. These are *coverage* failures: when the per-class seed set does not represent the full data distribution, purely local traversal never reaches some regions of the target, so the surrogate is left confidently wrong there. Diverse query generation is designed to reach exactly those unvisited regions. Both mechanisms are consistent with prior extraction results identifying near-boundary and diverse queries as the two most informative query types [14]; our aim is not to rediscover this, but to show *which explanation channel* lets an attacker realise each cheaply.

We instantiate a single *boundary-directed* extraction attack that extends the Autolycus traversal [10] with two components—*boundary search* and *diverse query generation*—and drive it with either explainer $\mathcal{E} \in \{\text{SHAP}, \text{LIME}\}$. The bin-edge stepping that constitutes LIME’s threshold channel (Phase 3) is Autolycus’s *own* mechanism, retained unchanged; our additions over Autolycus are the boundary search and the diverse generation, and the channel analysis itself. The two instantiations share the same scaffold and differ *only* where the decision *threshold* enters the computation (Phase 2 placement and Phase 3 stepping). This is deliberate: it turns the SHAP-vs-LIME comparison into a controlled experiment that isolates the threshold channel (H2), while ablating the explanation’s feature choice isolates the attribution channel (H1). Algorithm 1 summarises the three phases; we describe each below.

Algorithm 1 Boundary-directed explanation-guided extraction, instantiated with explainer $\mathcal{E} \in \{\text{SHAP}, \text{LIME}\}$.

Require: target API f , explainer $\mathcal{E}$, seed set $\mathcal{D}_0$, budget Q , top- k , step ϵ , confidence τ

Ensure: surrogate $\hat{f}$

- 1: $\mathcal{S} \leftarrow \mathcal{D}_0$; $\mathcal{T} \leftarrow \mathcal{D}_0$ $\triangleright$ traversal queue $\mathcal{S}$; labelled training set $\mathcal{T}$
- 2: $\mathcal{C} \leftarrow \text{DIVERSEGEN}(\mathcal{S}, \mathcal{E})$ $\triangleright$ plausibility-filtered or LIME-threshold
- 3: **for all** $x \in \mathcal{C}$ **do**
- 4: **if** $\max \hat{p}(x) \geq \tau$ **then**
- 5: $\mathcal{S} \leftarrow \mathcal{S} \cup \{x\}$
- 6: **end if**
- 7: **end for**
- 8: **for all** cross-class pairs (a, b) **do**
- 9: $c \leftarrow \text{blend top-}k \text{ features of } a \text{ toward } b$ $\triangleright$ LIME: snap to bin edge; SHAP: midpoint
- 10: $(c, M) \leftarrow \text{BISECT}(c, b)$ $\triangleright$ 8 steps; M = queried midpoints
- 11: $\mathcal{S} \leftarrow \mathcal{S} \cup \{c\}$; $\mathcal{T} \leftarrow \mathcal{T} \cup M$ $\triangleright$ recycle midpoints
- 12: **end for**
- 13: **while** budget remains **and** $\mathcal{S} \neq \emptyset$ **do**
- 14: $x \leftarrow \text{POP}(\mathcal{S})$; $\mathcal{T} \leftarrow \mathcal{T} \cup \{(x, f(x))\}$
- 15: select the top- k features of x by $\mathcal{E}$
- 16: **for all** selected feature i **do**
- 17: $x^\pm \leftarrow x$ with coordinate i stepped by $\pm \epsilon$ $\triangleright$ LIME: from the bin edge
- 18: add each valid, non-duplicate $x^\pm$ to $\mathcal{S}$
- 19: **end for**
- 20: **end while**
- 21: **return** $\hat{f} \leftarrow \text{TRAIN}(\mathcal{T})$

4.1 Phase 1: Diverse Query Generation

Autolycus seeds its traversal from a handful of per-class points and expands them by local perturbation, which tends to under-cover behavioural regions the seeds never reach. Phase 1 counters this with one of two generators, both returning *unqueried* candidates that are then queried and kept only if confidently classified ($\max \hat{p}(x) \geq \tau$, $\tau = 0.8$):

- **Plausibility-filtered** (explanation-free): evolutionary crossover and mutation of the seed pool, keeping only candidates within a nearest-neighbour distance percentile of the real data (a plausibility filter), then selecting by max–min spread. This is a purely geometric, explanation-free coverage baseline.
- **LIME-threshold** (explanation-informed): harvest the target’s LIME bin-edge thresholds *for free* from the already-queried seeds, synthesise seed-anchored candidates that cross those thresholds into different decision cells, plausibility-filter them, and greedily select the candidates covering the most distinct (feature, threshold, side) configurations. If the target surfaces no thresholds (e.g. a linear model), it falls back to the geometric generator.

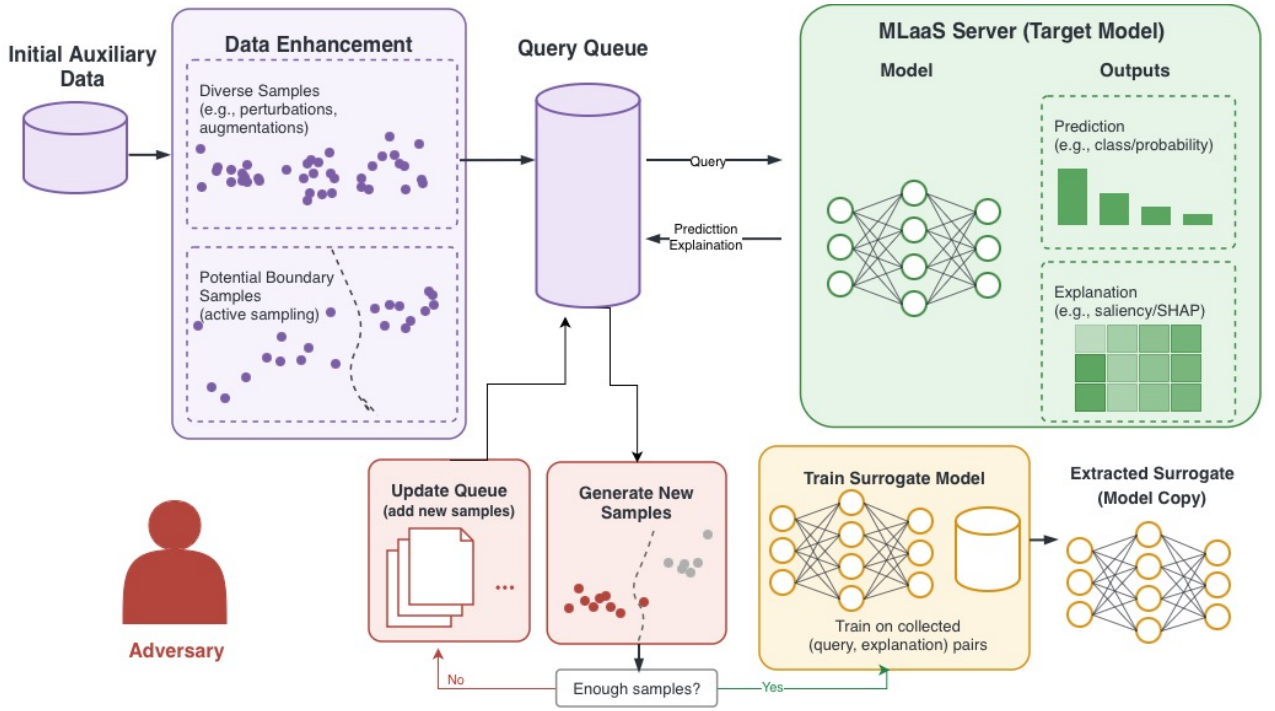


Figure 1: Explanation-guided model extraction pipeline. The adversary enhances a small auxiliary seed set with *diverse* and *boundary* samples, pushes candidates through a query queue to the explanation-augmented MLaaS API—receiving a prediction and a post-hoc explanation for each—uses those explanations to generate new queries, and trains a surrogate on the collected (query, prediction) pairs until the budget is spent. Our contribution is the data-enhancement stage (diverse generation and boundary search, §4.1–4.2); the explanation channels it consumes are the subject of our analysis (§5).

In its plausibility-filtered form Phase 1 uses no explanation information at all; the LIME-threshold variant additionally uses the threshold channel. Neither consults the attribution ranking.

4.2 Phase 2: Boundary Search

Phase 2 places training points *on* the decision boundary, where labels are most informative. For each cross-class pair (a, b) we build a child from a by blending its top- k features toward b , then run an 8-step bisection between the child and b to converge onto the boundary between their classes. The explainer governs *how* the child is placed, and this is precisely where the threshold channel enters:

- **LIME** snaps each continuous top- k feature to its local *bin edge* on the side toward b —a direct, free estimate of where the boundary crosses that coordinate.
- **SHAP**, having only magnitudes, places the blend at the *midpoint* and must rely on the subsequent bisection to *search* for the same boundary location.

Every bisection midpoint is queried against the target and therefore carries a label; we recycle all of them into $\mathcal{T}$ as boundary-concentrated training data (they are not ϵ -perturbed further).

4.3 Phase 3: Explanation-Guided Traversal

Phase 3 is the Autolytus-style expansion. We pop a sample x , query and label it, explain it with $\mathcal{E}$, select its top- k features (by $|\phi_i|$ for SHAP, by $|w_i|$ for LIME), and emit two children per feature stepped by $\pm\epsilon$; valid, non-duplicate children re-enter the queue until the budget is exhausted or all classes are sufficiently visited. We retain this expansion because it matters on its own—removing Phase 3 degrades fidelity, most sharply for tree targets—but the *attribution-based selection* is *not* what drives it: consistent with **H1**, replacing the explanation’s top- k choice with a random one leaves fidelity essentially unchanged on our primary (non-tree) targets (§6). What differentiates the two explainers here is, once more, the threshold channel: LIME steps $\pm\epsilon$ *from the feature’s bin edge*, so each step crosses the local threshold, whereas SHAP steps from the current value using only the (inert) attribution ranking. Phase 3 therefore contributes coverage and expansion; its sole explanation-supplied benefit is LIME’s threshold stepping, not the attribution ranking.

4.4 Query Accounting

Following Autolytus, the explanation is bundled with the prediction and *not* charged as a separate query; only the Phase 2 bisection intermediates consume additional budget. For a fair comparison we train the surrogate on *every* queried point—Phase-1 candidates, Phase-2 midpoints, and Phase-3 samples—so that Q counts exactly

the labelled points available to $\hat{f}$ and no paid-for query is discarded. Because the Phase-2 overhead can otherwise starve Phase 3 at small Q , a budget-scaling rule caps it to a fixed fraction of Q , relaxing at large budgets. These are accounting choices, not contributions. Throughout we report fidelity against the query *budget* Q rather than queries consumed, following the baseline’s convention; we return to what this does and does not favour in §6.2.

5 Analysis: Why Attribution Does Not Suffice

We set out the reasoning behind the two hypotheses of §3; §6 tests them.

Attribution is magnitude, not location (H1). An attribution vector $\phi(x)$ ranks features by local influence but says nothing about *where* along a feature the prediction changes. Two attackers handed the same ranking but stepping to different coordinates place entirely different queries, so the ranking alone cannot determine query placement. The ranking is moreover a *noisy* guide: although an explainer selects locally influential features more often than chance, replacing the explanation’s top- k choice with a random one leaves surrogate fidelity essentially unchanged (E2, §6). The attribution channel thus carries no signal the attacker can convert into fidelity beyond a random feature choice.

The threshold is the exploitable channel (H2). What an attacker can exploit is a *threshold*: a coordinate at which the prediction changes. LIME’s discretization discloses one for free—each returned (feature, bin) pair is a local threshold estimate—so a LIME attacker can place queries directly onto the boundary. SHAP, carrying only magnitudes, must *search* for that coordinate (e.g. by bisection), spending queries to recover what LIME reveals. This predicts that a threshold-bearing explainer beats a magnitude-only one on the identical scaffold, and that destroying the threshold—perturbing LIME’s bin edges—collapses the advantage; both are tested in E3 (§6). With H1, this recasts explanation leakage as a matter of decision *thresholds*, not attribution or geometry.

6 Experiments

6.1 Setup

We evaluate on six public tabular datasets (Table 1) covering continuous, categorical, and mixed feature schemas, against five target families: logistic regression (LR), naive Bayes (NB), k -NN, decision tree (DT), and random forest (RF). We scope to tabular data throughout, following the attack we analyse [10]. Following the Autolycus convention, the LIME attacker uses $n=1$ seed per class and explores $k=3$ top features, and the SHAP attacker uses $n=5$ and $k=3$. Query budgets are per dataset (Table 1); following Autolycus we plot fidelity against the query *budget* rather than queries consumed (§4.4).

Compared arms always run on *paired* seed sets (identical draws, identical RNG state), and surrogate fidelity (Eq. 1) is aggregated by the *mean* over refits. Significance is a paired Wilcoxon signed-rank test over the 10 paired observations, with $*p<.05$ and $**p<.01$. With 10 pairs the smallest attainable two-sided p -value is $2/2^{10} \approx .002$, reached only when every pair falls the same way; $**$ therefore marks a near-unanimous effect across target instantiations rather than an arbitrarily small p .

Table 1: Datasets. n is the total size; *cont.* counts continuous features (the rest are categorical, for which LIME’s discretization yields few distinct cut points). Train / eval / aux follow the 75 : 15 : 10 split of the attack we analyse: the target is fit on train, fidelity is measured on eval, and the adversary’s auxiliary sample is drawn from aux. Evaluation-set size sets the resolution of every per-cell number: one held-out point is 0.06pp on pendigits but 1.18pp on breast, so effects of a few points are resolvable on the former and not on the latter.

	n	feat.	cont.	cls.	train	eval	aux	
crop	1,700	7	7	17	1,275	255	170	1000
adult	32,561	11	3	2	24,420	4,884	3,257	1000
breast	569	30	30	2	426	85	58	100
nursery	12,960	8	0	3	9,720	1,944	1,296	1000
mushroom	8,123	21	0	2	6,092	1,218	813	1000
pendigits	10,992	16	16	10	8,244	1,648	1,100	500

Protocol. We deviate from the original attack’s reporting in two ways, both toward more conservative estimates. *Surrogate aggregation:* the original trains R surrogate refits per arm and reports the single one most similar to the target—a quantity measured on the evaluation set, so the selection is made using the reported metric. We instead report the mean over the R refits, which is unbiased. (R denotes surrogate refits and is unrelated to the top- k feature budget k ; we use $R=20$ for DT, $R=10$ for RF, and $R=1$ otherwise.) *Target instantiation:* the original varies only the attacker’s seed sets against a single fitted target, so its reported spread reflects seed noise rather than variation across targets. For the ablations (§6.4, §6.5) we therefore draw 10 independent train/test splits and retrain the target on each, pairing all arms within a split, so each observation is an independent target instantiation. We deliberately retain the original single-split protocol for the attack comparison of §6.2, so that those numbers remain comparable to the published baseline. [TODO: I am not sure if I want to give this as the main result. maybe I’ll give our version of comparison in the main theirs in the appendix so put a mark on this sentence, we will decide later]

6.2 E4: The attack used as the measurement instrument

Figure 2 plots surrogate fidelity against query budget for our LIME attack and the Autolycus LIME baseline. At a single fixed diverse budget, our attack improves fidelity on 18 of 25 target/dataset combinations (10 significant at $p<.05$, against 2 significant regressions), with the most consistent gains on the non-tree targets that have headroom (LR/NB on crop, nursery, and mushroom).

Part of this margin is an accounting effect worth stating plainly. Autolycus’s traversal terminates when its queue empties, which on several targets happens well before the budget is exhausted; its curve then flattens because it *cannot* spend the remaining budget, not because additional queries would not help. Our Phase-1 and Phase-2 components keep generating admissible queries, so they continue to convert budget into fidelity. Read at equal *budget* the comparison therefore flattens our attack, and read at equal *consumed queries* it would flatter Autolycus; we plot budget because that is the

baseline’s convention and the quantity a defender actually meters. The $k=3$ control in Appendix A, where the baseline does spend its full budget, isolates how much of the margin is saturation rather than better placement.

The same attack helps under SHAP. Figure 3 repeats E4 with SHAP as the explainer ($n=5$, $k=3$): our SHAP3 attack improves over Autolycus-SHAP on the same non-tree targets (LR/NB/KNN, largest where the baseline has headroom, e.g. nursery and mushroom), while tree targets sit near their fidelity ceiling and show no gain. That the same boundary-and-diversity machinery helps under both explainers—independent of the particular attribution scores it consumes—is the first indication that the attribution channel is not what drives the gain (H1); the LIME-specific advantage of a free *threshold* is isolated separately in E3.

Diverse-budget sensitivity. The diverse generation (Phase 1) is regime-dependent: its optimal budget is roughly bimodal, helping monotonically on high-headroom targets (best at the maximum budget, e.g. NB/adult: 0.61→0.96 from no-diverse to full) and hurting on near-ceiling or small continuous targets (best with no diverse). Because the budget is not the contribution, the headline figure fixes it to a single value (cap=15, the best global choice: +3.4pp mean, 18/25 wins) rather than tuning per target. Fixing it this way costs accuracy on the targets that prefer a different budget, and we prefer that cost to a per-target tuning that would not be available to a real attacker.

6.3 E1: Which part of the attack drives the gain?

Our attack adds two components to the Autolycus traversal: diverse query generation (Phase 1) and boundary search (Phase 2). We isolate their contributions with three nested arms on the same seed sets—Autolycus alone (*base*), +boundary (Phase 2, no diverse), and +diverse on top (the full attack)—so that $\text{boundary} = (\text{+boundary}) - \text{base}$ and $\text{diverse} = \text{full} - (\text{+boundary})$. Figure 4 reports the result.

The two components contribute about equally on average (+1.6 and +1.8pp), but *which* one drives the gain is strongly target-dependent: naive Bayes benefits almost entirely from *diverse generation* (+6.4pp)—a generative model needs distribution coverage—whereas random forest and logistic regression benefit almost entirely from *boundary search* (+4.7, +2.5pp), which places labelled points on the decision surface. In every case the gain is query *placement*, not the attribution ranking: consistent with H1, the components determine *where* queries land, and (E2) replacing the explanation’s feature ranking with a random one does not systematically change fidelity. One caveat bounds this decomposition: the boundary term as measured here bundles Phase-2 boundary search with the Phase-3 threshold stepping it inherits from Autolycus, so it should be read as the combined contribution of both rather than of bisection alone. §6.5 isolates the threshold component directly.

The split is also strongly *dataset*-dependent (Figure 5): diverse generation carries almost the entire gain on ADULT/NB (+34pp, a generative target starved of coverage), while boundary search dominates on the tree targets of CROP and the linear targets of BREAST. Neither component is tied to a particular dataset—both

are query *placement* mechanisms—which is why the aggregate in Figure 4 splits evenly.

6.4 E2: Is the attribution channel the leak? (H1)

We replace the explanation’s top- k feature selection with a random- k choice, holding the rest of the attack fixed (the *feature* axis of the base-Autolycus 2×2), and measure the change in surrogate fidelity. If the attribution ranking carries exploitable signal, random selection should be worse. Across both explainers the attribution channel is *weak and inconsistent* ($S=10$ paired sets):

- **SHAP: no net advantage.** On the base Autolycus attack, SHAP-guided selection is *no better on average* than random no-explanation traversal (Table 2; mean +0.02pp over 30 cells, with 5 significantly positive and 6 significantly negative). Where it helps it is confined to *tree* targets on datasets with a few dominant features, for which the *exact* TreeSHAP pinpoints true splits (MUSHROOM DT +9.8, RF +6.9; PENDIGITS DT +4.1pp, all $p < .01$). But the same explanation that helps trees on MUSHROOM *hurts* logistic regression on that very dataset (−11.2pp), and it misleads across the board on NURSERY (LR −10.2, NB −9.1, KNN −4.6pp). The sign of the effect is a property of the target, not of the explanation, so an attacker choosing to consult SHAP has no way to know whether doing so will help or harm.
- **LIME: no systematic advantage either.** Replacing LIME’s ranking with a random choice—holding the threshold available to both arms—changes fidelity by a mean of +0.6pp and a median of −0.3pp, with significant effects in *both* directions (Table 3). This is weaker than claiming the channel is inert: individual cells show real effects, but their sign is not predictable from the dataset or the target family, so an attacker gains nothing it can rely on. The instability is itself informative: individual cells move by several points under changes that affect only *which* features the control happens to draw, which is what one expects when the quantity being measured is close to zero.

So neither explainer’s attribution ranking confers a systematic extraction advantage. This is not merely an absence of significance: the contrast with the *threshold* channel is *directional*. Over the identical cells, threshold ablation is significantly positive eleven times and significantly negative *never* (mean +6.8pp), whereas attribution is significant in both directions around a mean of +0.6pp (§6.5). One channel has a sign; the other does not. What LIME supplies is not a useful feature ranking but a decision *threshold*.

6.5 E3: Is the threshold (bin edge) the leak? (H2)

We now isolate the *threshold* axis of the same base-Autolycus 2×2 : with the feature choice fixed to LIME’s ranking, we perturb from the bin edge LIME exposes (default) versus from the sample’s current value (no-threshold), everything else identical. Because the base attack already snaps to the bin edge and has *no* boundary search to mask the effect, this is the cleanest possible test of H2.

The threshold channel is directionally consistent. On the two datasets whose evaluation sets can resolve it, the bin-edge signal

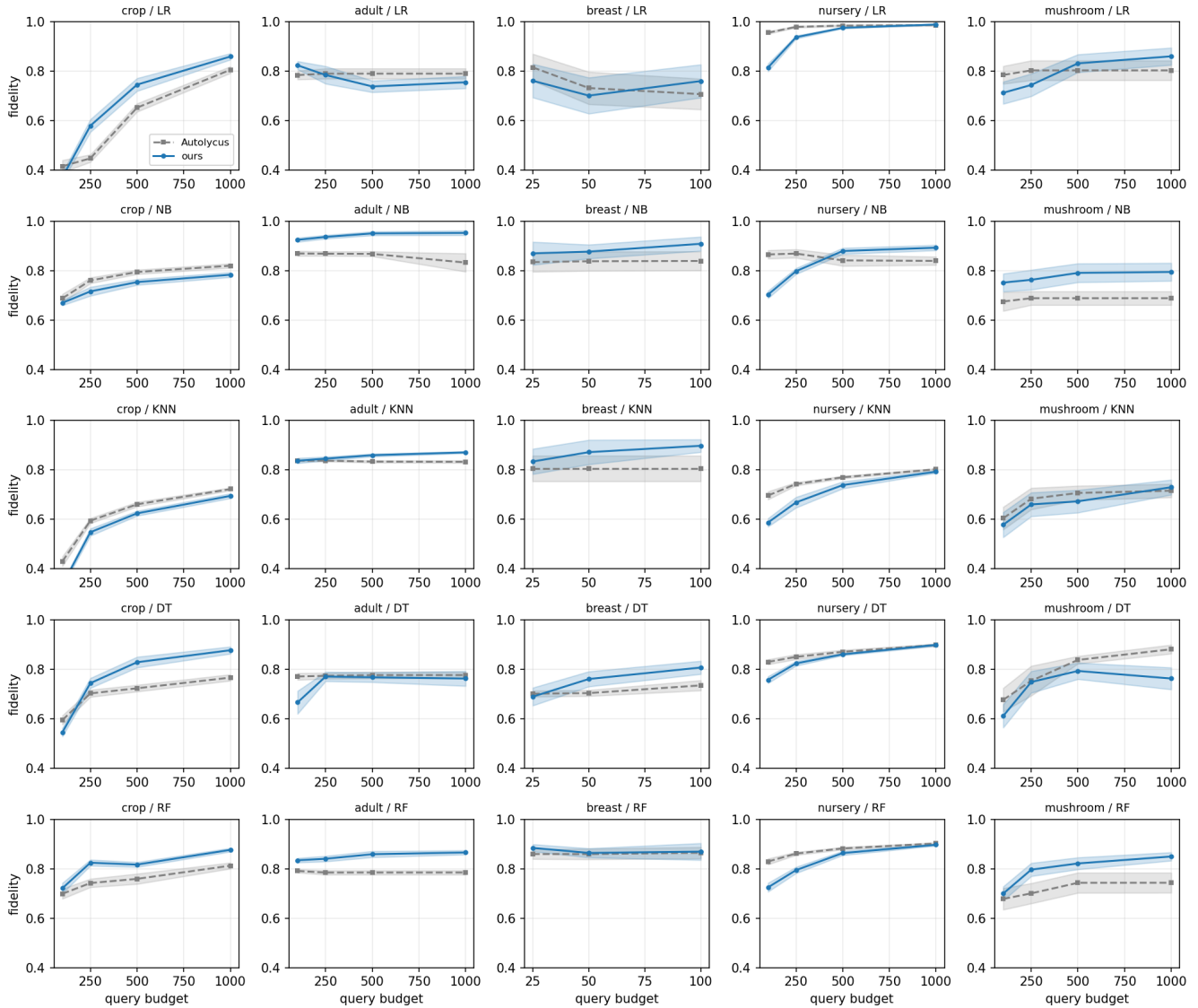


Figure 2: E4: surrogate fidelity vs. query budget—our LIME attack (solid, blue) vs. Autolycus-LIME (dashed, grey), per target (rows) and dataset (columns). Shaded band is the standard error over the $S=10$ seed sets. Ours uses a single fixed diverse budget (cap=15, not tuned per target); $n=1$, $k=3$ (breast $Q \leq 100$, others $Q \leq 1000$). All panels share a fixed y -axis (0.4–1.0) for cross-panel comparability.

produces large, significant gains: crop (agronomy, +15 to +41pp) and pendigits (pen-trajectory, +5 to +11pp), across ten independent target instantiations. The asymmetry is what matters: over these cells the threshold ablation is significantly positive eleven times and significantly negative *never*, whereas the attribution ablation on the identical cells is significant in both directions (Table 3). This is precisely the signal *absent* from SHAP: on pendigits, LIME’s threshold yields +5 to +11pp while SHAP’s attribution yields +0.7pp (n.s.), because Shapley magnitudes carry no “where” information. H2 holds where H1 fails.

Scope—the mechanism predicts where the channel is weak. The threshold channel is load-bearing only when LIME’s per-feature bin edge is a meaningful proxy for the decision boundary *and* the base attack has room to improve. Both conditions bite. Where features are *categorical*, discretization yields few distinct cut points and the channel is correspondingly weak and unstable: on NURSERY the effect is at most a few points against +15 to +41pp on crop, and it does not reproduce reliably—the NB cell measures +5.0 in one run and +1.0 in another differing only in random seed. We report this rather than a clean null because NURSERY’s features are label-encoded ordinals, so quartiles do fall on real level boundaries; but

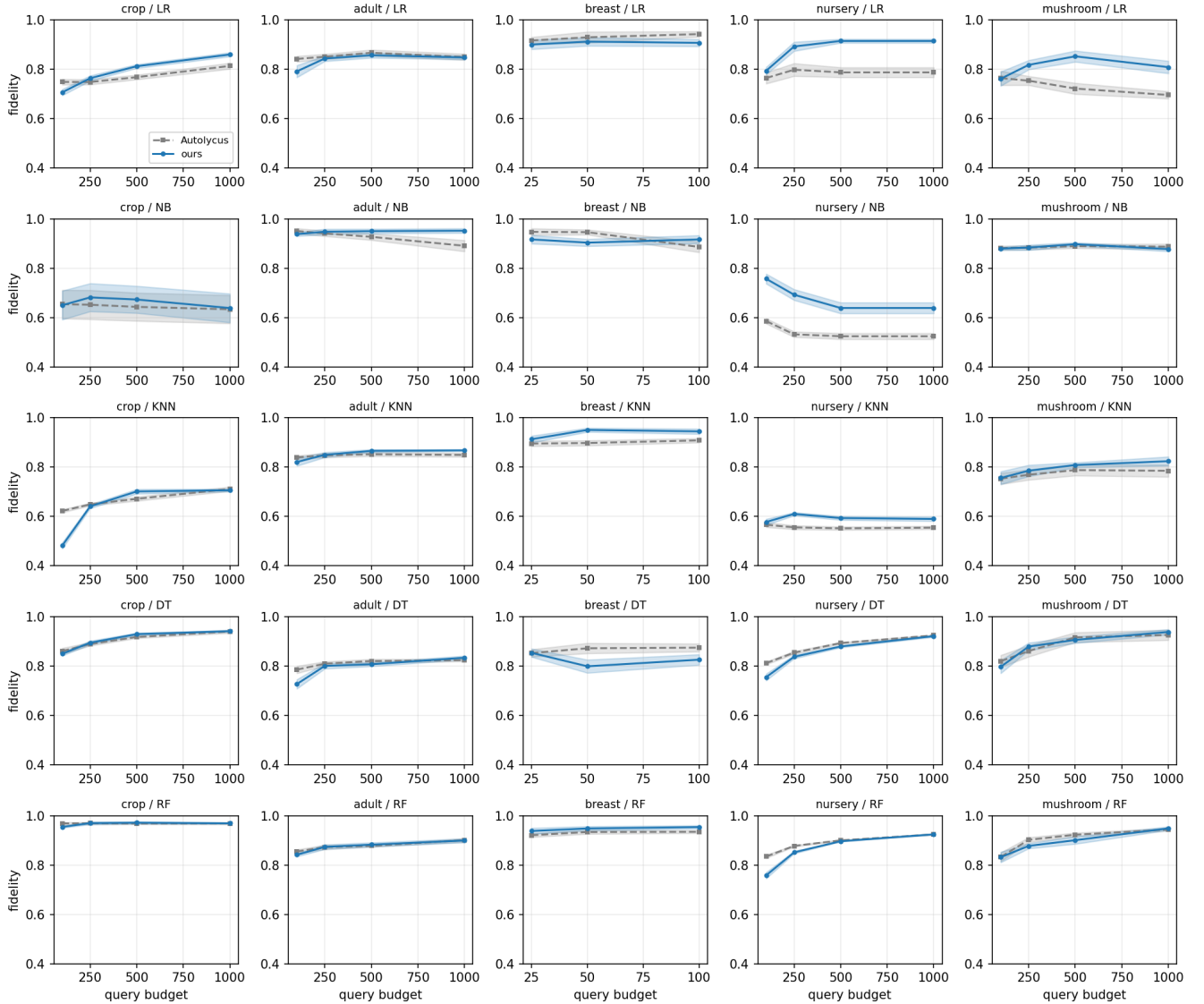


Figure 3: SHAP companion to E4: surrogate fidelity vs. query budget—our SHAP3 attack (solid, blue) vs. Autolycus-SHAP (dashed, grey), per target (rows) and dataset (columns). Shaded band is the standard error over the $S=10$ seed sets; $n=5$, $k=3$. All panels share a fixed y -axis (0.4–1.0) for cross-panel comparability.

the channel there is too small to support a claim in either direction. Where the target is *near-ceiling* (breast) there is no headroom for any channel to exploit. We scope all experiments to tabular data, following the attack we extend; we therefore make no claim about high-dimensional or non-axis-aligned regimes.

6.6 E5: Is the threshold privileged to the target?

§6.5 establishes that the discretization threshold is the channel that matters. It does not establish that the threshold is *information the service discloses*. LIME’s default `QuantileDiscretizer` places its bin edges at the 25/50/75 percentiles of the explainer’s background data—a statistic of the data distribution that is independent of the

model and of the labels. If an adversary can estimate those percentiles from data it already holds, the discretization accelerates extraction without disclosing anything.

We test this directly on LIME itself. Holding the traversal, the seeds, and the RNG fixed, we vary only the data the discretizer is fit on, across three paired arms: `nothresh` (no bin edge), `tgt` (discretizer fit on the target’s X_{train} —what the service exposes), and `aux` (discretizer fit on the adversary’s own auxiliary data). We report thr_{tgt} and thr_{aux} , each measured against `nothresh`, and the privileged advantage leak = $\text{tgt} - \text{aux}$.

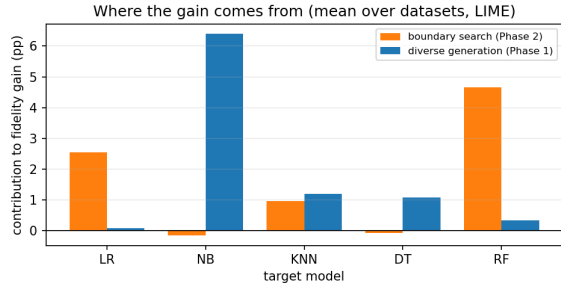


Figure 4: E1: contribution of each component to the LIME fidelity gain over Autolytus (percentage points, mean over datasets, top budget). Boundary search dominates for RF/LR; diverse generation dominates for NB.

Table 2: E2 on the *base* Autolytus SHAP attack: fidelity gain (percentage points) of SHAP-guided feature selection over random no-explanation traversal ($n=5$, $k=3$; $*p<.05$, $p<.01$). Multi-split protocol: 10 independent train/test splits with the target retrained and the SHAP explainer refit on each, arms paired within a split. Positive = SHAP helps. Significant effects occur in both directions and their sign tracks the *target*, not the explanation: SHAP aids trees on MUSHROOM while simultaneously harming LR on the same dataset, and misleads across NURSERY.**

	crop	adult	breast	nursery	mushroom	pendigits
LR	-5.5*	-0.0	+1.2	-10.2**	-11.2**	+1.3
NB	-1.0	+2.5	+1.4	-9.1**	+3.8	+0.5
KNN	+1.0	+1.0	-0.4	-4.6**	+3.6*	+0.7
DT	+1.3	+0.8	+2.3	-1.1*	+9.8**	+4.1**
RF	+0.4	+0.7	+0.3	+0.7*	+6.9**	-0.6

The discretization helps, but it is not a disclosure. Across all twelve cells in which the threshold channel is significantly positive—spanning five datasets and every target family—the adversary’s own discretizer reproduces the gain (Table 5): on crop, thr_{aux} reaches +41.6pp against thr_{tgt} ’s +39.6pp, and no leak term is statistically significant—the differences average -0.4pp and span -3.1 to +2.2pp, against threshold effects of up to +39.6pp. The sign is inconsistent, the self-computed grid being marginally better in seven of the twelve cells, which is what one expects of two estimates of the same quantity rather than of a degraded substitute. The result covers continuous (crop, pendigits), near-ceiling (breast), and categorical (nursery) targets alike; if anything it is cleanest on categorical data, where quantiles over a handful of ordinal levels are estimable from almost any sample. The threshold channel is therefore a *self-computable* property of the data distribution rather than privileged information about the model. This reframes the finding of §6.5: what LIME’s discretization supplies is a data-adaptive grid of candidate cut points that makes boundary-straddling queries cheap—valuable to an attacker, but not something the service is uniquely able to reveal.

Table 3: E2 on the *base* Autolytus LIME attack: fidelity gain (percentage points) from selecting the top- k features by LIME’s ranking rather than at random, with the bin edge available to both arms ($n=1$, $k=3$; $*p<.05$, $p<.01$). Multi-split protocol: 10 independent train/test splits with the target retrained on each, arms paired within a split. Because LIME fits a *sparse* local model and does not score every feature, the random arm recovers the missing bin edges from the explainer’s own discretizer, so the two arms differ only in *which* features are perturbed and not in whether a threshold is available. $\dagger$ marks cells where more than four of the ten splits produced identical fidelity in both arms, leaving too few non-tied pairs for a meaningful test. Effects are small and inconsistent in sign—compare the *threshold* ablation on the identical cells (Table 4).**

	crop	pendigits	breast	adult	nursery	mushroom
LR	-3.8*	-0.8	+6.0	-4.5	-0.2	
NB	+6.8**	-1.9	+5.2	-1.8 $\dagger$	-6.0	
KNN	+4.4	-0.1	+4.0 $\dagger$	-0.3	+5.7**	
DT	-2.2	-4.4	-2.7	-6.0	+1.5	

Table 4: E3 on the *base* Autolytus LIME attack: fidelity gain (percentage points) from perturbing at the exposed *bin edge* vs. the current value, features held to LIME’s ranking ($n=1$, $k=3$; $*p<.05$, $p<.01$). Same protocol, cells, and layout as Table 3, so the two channels are directly comparable. Positive = the threshold helps. Effects are large and significant on the well-measured continuous datasets (crop, pendigits) and are directionally consistent: no cell is significantly negative. WINE and BREAST are reported for completeness but their evaluation sets are small (27 and 85 samples, i.e. 3.70 and 1.18pp per sample), so they cannot resolve effects of this size.**

	crop	pendigits	breast	adult	nursery	mushroom
LR	+40.9**	+9.2*	-10.2	-1.8	-0.8	-8.7
NB	+24.9**	+5.1**	+11.2*	+0.0	+5.0**	-0.2
KNN	+15.2**	-0.0	-0.2	-0.3	+1.0**	+13.9**
DT	+38.0**	+11.1**	+4.8	+2.6	+0.6	+2.1

How much auxiliary data this requires. Self-computation is bounded by the quality of the adversary’s distributional estimate, and the threat model we inherit permits a range of auxiliary set sizes [10]. Fitting the discretizer on the full auxiliary partition (the “comprehensive” end of that range) gives the result above. Restricting it to the $n=1$ -per-class seed set (the minimal end, and the setting the original evaluation uses) preserves the result on crop, where all leak terms remain insignificant, but degrades it on pendigits, where ten samples must estimate quartiles over sixteen wide-range continuous features and one leak term becomes significant (+3.0pp, $p<.05$). A third variant, re-estimating the bins online from the accumulated query set, performs *worse* than either: the attack deliberately concentrates queries near decision boundaries, so its own query history is a biased estimator of the marginal distribution the bins

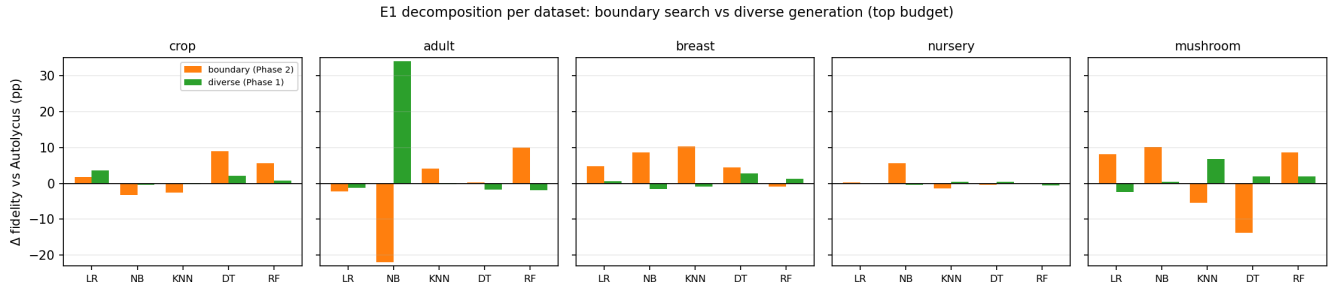


Figure 5: E1, per dataset: contribution of boundary search (Phase 2, orange) and diverse generation (Phase 1, green) to the LIME fidelity gain over Autolycus (percentage points, top budget, mean over $S=10$ seed sets), one panel per dataset, grouped by target model. Shared y -axis; the ADULT/NB diverse bar (+34pp) sets the scale.

Table 5: E5: the threshold channel is self-computable. Every cell in which the threshold channel significantly helps the attack ($\text{thr}_{\text{tgt}} > 0$, $p < .05$), across all five datasets, ordered by effect size. thr_{tgt} is the gain from the target’s discretizer and thr_{aux} the gain from one the adversary fits on its own auxiliary sample, both measured against a no-threshold baseline; leak is their difference, i.e. the advantage attributable to the target’s data. Multi-split, $n=1$. No leak term is significant, and the largest is 3.1pp against threshold effects of up to 39.6pp. We restrict the table to cells with a real effect because a leak term is uninformative where there is nothing to reproduce.

dataset	model	thr_{tgt}	thr_{aux}	leak
crop	LR	+39.6	+41.6	−2.0
crop	DT	+37.4	+37.4	+0.0
crop	NB	+25.0	+24.3	+0.6
crop	RF	+22.7	+22.4	+0.3
crop	KNN	+15.1	+18.2	−3.1
mushroom	KNN	+14.0	+16.5	−2.5
breast	NB	+11.2	+11.9	−0.7
pendigits	DT	+11.1	+9.0	+2.2
pendigits	LR	+9.2	+9.1	+0.1
pendigits	NB	+5.1	+5.1	−0.0
nursery	NB	+5.0	+4.7	+0.3
nursery	KNN	+1.0	+1.3	−0.3

are meant to summarise. The adversary’s advantage thus rests on holding a modestly sized representative sample—not on anything the explanation uniquely reveals.

Why obfuscating the channel is unlikely to defend it. Because the channel localises to LIME’s `discretize_continuous` step, disabling or coarsening it is the natural candidate mitigation. Two caveats prevent us from claiming it as a validated defense. First, E5 argues it would be *insufficient*: an adversary holding a representative auxiliary sample can reconstruct an equivalent grid, so withholding the discretization removes a convenience rather than the capability. Second, preliminary attempts to perturb the bins rather than remove them (more bins, rounded edges) did not reduce

extraction. Measuring what a defender actually gains by withholding or distorting the discretization, against an adversary that adapts by self-supplying it, is left to future work.

6.7 E6: Does the adversary need the explanation at all?

E2 and E5 each remove one channel. Composing them gives the question a defender actually cares about: how much worse off is an adversary that reads *nothing* from the explanation? We compare three arms of the base attack, paired within splits as before: `full` (the target’s feature ranking and the target’s bin edges—the real service), `randaux` (features chosen at random and bin edges from a discretizer the adversary fits on its own auxiliary sample—no explanation content at all), and `randnoin` (random features and no grid, the floor). Two quantities follow: `gap = full − randaux`, the residual value of explanation access, and `grid = randaux − randnoin`, the value of the self-computed grid alone.

The grid matters; its source does not. The self-computed grid is worth a great deal: +44.4pp on crop/LR and +40.0pp on crop/DT, significant in eleven of twenty cells, and it costs the adversary nothing beyond the auxiliary sample the threat model already grants it. Against that, the target’s own explanation is worth +0.4pp on average, with four significantly positive cells and two significantly negative—significance in both directions around approximately zero, the same signature we saw for the attribution channel in §6.4. An adversary that reads nothing from the explanation is therefore not materially worse off than one that reads all of it.

This is the sharpest form of the paper’s claim, and it is worth being clear about what it does and does not say. It does *not* say that query placement is unimportant: the grid is the single largest effect anywhere in our experiments. It says that the placement information an explanation supplies is a statistic of the data distribution, which an adversary holding a modest sample from that distribution can estimate for itself. The service’s contribution is convenience, not disclosure. Notably this holds on the *base* attack, without the query-synthesis machinery of §4: the conclusion does not depend on our additions.

Table 6: E6: what the explanation is still worth once the adversary self-supplies a grid. gap is the target’s explanation minus a self-supplied substitute; grid is what the self-computed grid alone buys over no grid. Base attack, multi-split, $n=1$; $*p<.05$. The grid is worth up to +44pp, while the explanation adds +0.4pp on average with significant effects in both directions.

dataset	model	full	gap	grid
crop	LR	78.8	-4.5	+44.4*
crop	NB	80.7	+5.9*	+17.8*
crop	KNN	66.6	+3.8*	+9.6*
crop	DT	80.6	+2.0	+40.0*
crop	RF	81.0	-4.3	+20.4*
breast	LR	54.9	-6.9	-6.2
breast	NB	85.1	+6.7	+4.3
breast	KNN	79.7	+13.1*	-2.5
breast	DT	68.7	+0.8	+3.6
breast	RF	76.8	-4.2	-1.1
nursery	LR	98.0	-0.2	-0.1
nursery	NB	83.7	-6.5*	+5.6*
nursery	KNN	81.2	+7.0*	+6.5*
nursery	DT	89.7	+0.6	+2.0
nursery	RF	91.0	+1.0	+3.8*
pendigits	LR	62.6	+1.0	+6.4
pendigits	NB	71.4	-0.5	+4.4*
pendigits	KNN	62.3	+0.1	-0.2
pendigits	DT	42.1	-0.7	+9.0*
pendigits	RF	49.4	-6.0*	+3.4*
mushroom	—	—	—	—

[TODO: [AC] Pending runs, to fold in here: (i) E5 coverage for nursery + mushroom (breast done: NB leak -0.5, n.s.); (ii) the aux-size sensitivity (seed-set vs full pool) and the online-refit negative result, currently only in notes.]

7 Discussion and Limitations

Where the leak lives. Our decomposition recasts explanation-guided extraction as a question about *decision thresholds*. The exploitable signal is not the attribution magnitude (§6.4) nor, as prior framings imply, the boundary *geometry*, but the discretization threshold LIME publishes for free (§6.5). SHAP, which reports magnitude without a threshold, carries no comparable signal. The practical consequence is asymmetric: an MLaaS provider can serve SHAP-style magnitude attributions at substantially lower extraction risk than LIME-style discretized explanations.

What kind of information the threshold is. LIME’s default bin edges are the 25/50/75 percentiles of the explainer’s background data—a property of the data distribution, not of the model or its labels. The channel we identify is therefore best read as a data-adaptive grid of candidate cut points that lets the attacker place boundary-straddling queries cheaply, rather than as a disclosure of the model’s own decision surface. Whether an adversary can substitute a grid estimated from its own auxiliary data—in which

case the discretization is a convenience rather than a disclosure—the natural next question, and we regard it as open.

Scope and limitations. The threshold channel is not universal, and we do not claim it is. It is load-bearing only for continuous tabular targets with fidelity headroom (crop, pendigits); on categorical features it is an order of magnitude weaker, since discretizing a few ordinal levels yields few usable cut points, and on near-ceiling targets it is absent because the attack is already saturated. We scope to tabular data throughout, following the attack we extend, and make no claim about high-dimensional or non-axis-aligned domains.

Two measurement caveats bound what our per-cell numbers can support. First, *evaluation-set size* governs resolution: fidelity on WINE is measured over 27 held-out points (3.70pp per sample) and on BREAST over 85 (1.18pp), against 255 for crop and 1648 for pendigits. Effects on the two small sets are not resolvable at the magnitudes of interest, and we therefore rest no claim on them—a caution that applies equally to prior work reporting per-cell fidelity on small splits. Second, our threshold ablation does not run LIME×RF at full budget (traversal cost), so the ensemble case is not covered. Finally, the crossing-rate objective in Phase 1 is a proxy for decision-cell coverage.

8 Conclusion

Decomposing an explanation-guided extraction attack into an attribution channel and a threshold channel shows that the attribution ranking—in both LIME and SHAP—confers no systematic advantage, being centred on zero and inconsistent in sign. The one channel that reliably helps is the decision threshold LIME’s discretization exposes, worth up to 41 fidelity points and directionally consistent. That threshold, however, is a quantile of the data distribution rather than a property of the model, and an adversary holding a representative auxiliary sample reconstructs an equivalent grid with no measurable loss. What an explanation contributes to extraction is therefore query-placement convenience, not disclosure—which also means that obfuscating the channel is unlikely to defend it. Characterising what a defender can actually withhold from an adaptive adversary remains open.

A Robustness to seed count and feature budget

The main experiments use LIME at $k=3$, $n=1$ (the Autolycus setting). We report here two alternative settings that vary the number of perturbed top features (k) and seeds per class (n), under the identical protocol and fixed diverse budget of the main text.

Figure 6 ($k=1$, $n=3$) shows the largest apparent margin over Autolycus, but this is an artefact of *base saturation*: at $k=1$ Autolycus’s traversal exhausts its queue and cannot spend the budget—its curves go flat—while our diverse generation keeps placing queries. Figure 7 ($k=3$, $n=3$) is the fair control: more seeds, but Autolycus also uses the full budget. There the mean margin is *smaller* than at $n=1$ (+1.9 vs +3.4pp), because additional seeds help Autolycus as much as us. We therefore keep $k=3$, $n=1$ as the headline; the larger $k=1$ margin reflects Autolycus’s budget under-utilisation, not a seed advantage of our method.

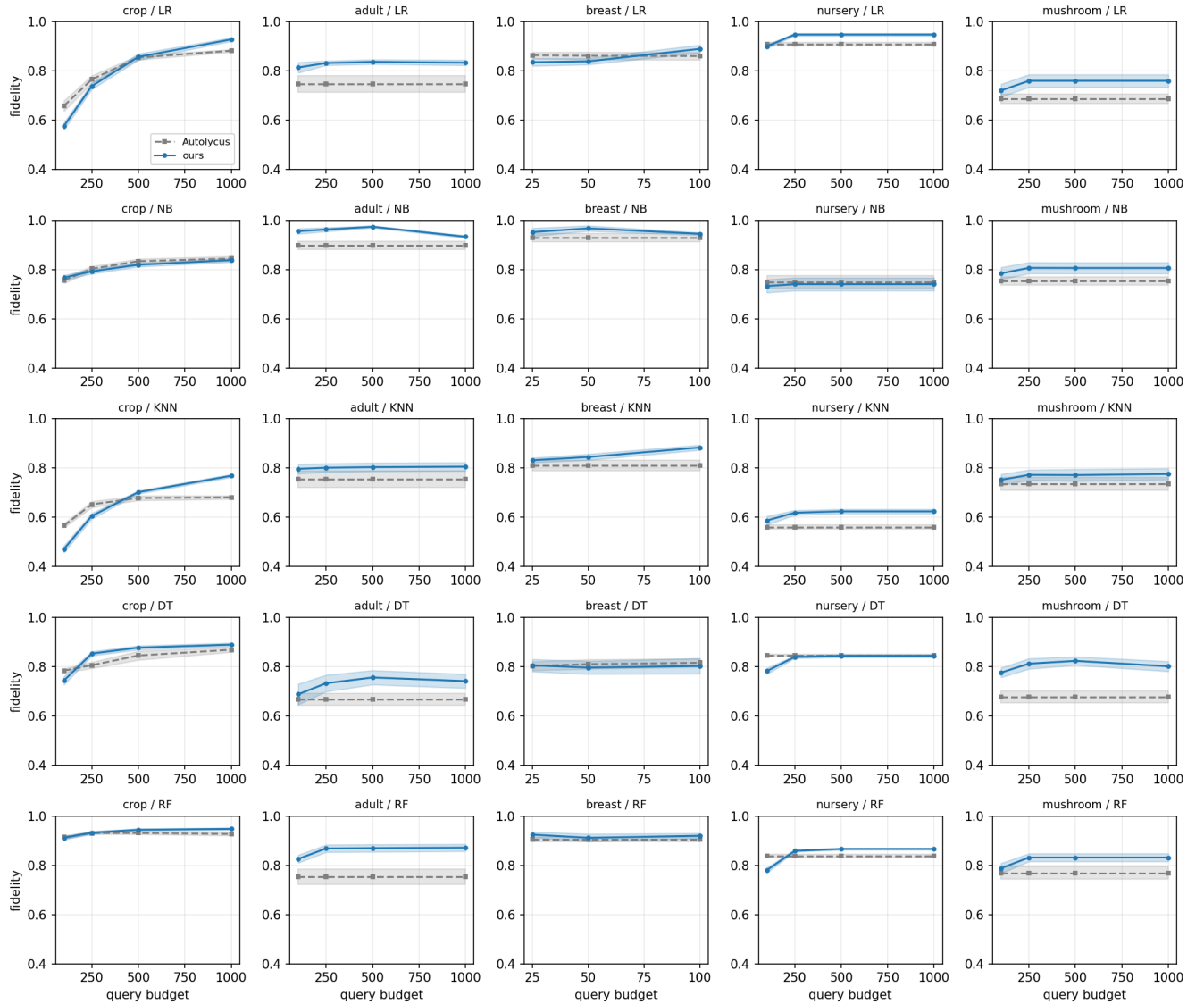


Figure 6: LIME at $k=1$, $n=3$ (secondary). Fidelity vs. query budget, ours (blue, solid) vs. Autolytus (grey, dashed); band = standard error over 10 sets. Autolytus’s curves flatten early—its traversal saturates at $k=1$. Shared fixed y -axis (0.4–1.0).

Ethical Considerations

This work studies an *attack*—model extraction guided by post-hoc explanations—in order to characterise a concrete privacy risk of explainable machine-learning services. All experiments use publicly available benchmark datasets and target models that we train ourselves; no proprietary system, personal data, or live service was queried, and no human subjects were involved. The techniques extend already-published extraction attacks and are evaluated in a controlled setting, so we assess the incremental disclosure risk as low. Our findings are, on balance, *risk-reducing*: we show that the model-specific content of an explanation contributes little to extraction, and that the component which does contribute is one an adversary can reconstruct without the service’s help. This bounds

the incremental risk that publishing an explanation creates, rather than expanding it. The work was not submitted to an external ethics board, as it involves no human-subjects or live-system experimentation.

Open Science

We will release our attack implementation and evaluation harness, together with scripts to reproduce every figure and table, upon acceptance. All datasets used are already public.

AI Use

AI-based tools were used to assist with drafting and revising the text (improving flow and correcting phrasing), to help search for

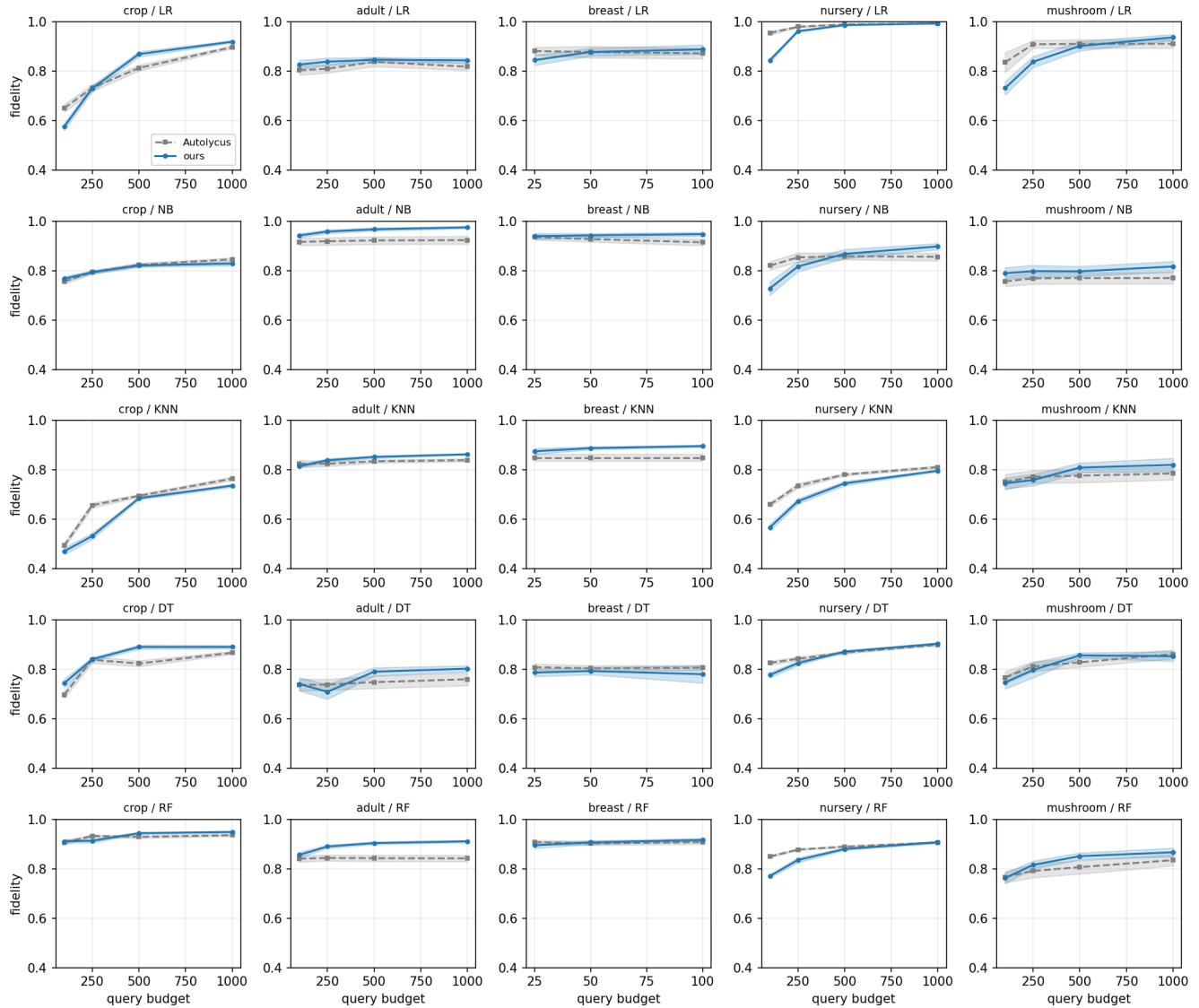


Figure 7: LIME at $k=3$, $n=3$ (fairness control). With $k=3$ Autolytus uses the full budget, and our margin over it is smaller than at $n=1$ —confirming the larger $k=1$ margin is base saturation, not a seed benefit. Shared fixed y -axis (0.4–1.0).

related work, and for pre-submission review. The authors have manually verified, and are responsible for, the accuracy, originality, and integrity of all content, including every technical claim, experiment, and citation.

References

- [1] Ulrich Aivodji, Alexandre Bolot, and Sébastien Gambs. 2020. Model Extraction from Counterfactual Explanations. *arXiv:2009.01884* (2020).
- [2] Jacson Rodrigues Correia-Silva, Rodrigo F Berriel, Claudine Badue, Alberto F De Souza, and Thiago Oliveira-Santos. 2018. Copycat CNN: Stealing Knowledge by Persuading Confession with Random Non-Labeled Data. In *IJCNN*.
- [3] Fatima Ezzeddine, Osama Zammar, Silvia Giordano, and Omran Ayoub. 2026. Explanations Leak: Membership Inference with Differential Privacy and Active Learning Defense. *arXiv:2602.03611* (2026).
- [4] Matthew Jagielski, Nicholas Carlini, David Berthelot, Alex Kurakin, and Nicolas Papernot. 2020. High Accuracy and High Fidelity Extraction of Neural Networks. In *USENIX Security*.
- [5] Pratik Karmakar and Debabrota Basu. 2023. Marich: A Query-efficient Distributionally Equivalent Model Extraction Attack. In *NeurIPS*.
- [6] Awa Khouna, Julien Ferry, and Thibaut Vidal. 2025. From Counterfactuals to Trees: Competitive Analysis of Model Extraction Attacks. In *NeurIPS*. *arXiv:2502.05325*.
- [7] Scott M Lundberg and Su-In Lee. 2017. A Unified Approach to Interpreting Model Predictions. In *NeurIPS*.
- [8] Smitha Milli, Ludwig Schmidt, Anca D Dragan, and Moritz Hardt. 2019. Model Reconstruction from Model Explanations. In *FAccT*.
- [9] Thanh Tam Nguyen, Thanh Trung Huynh, Zhao Ren, Thanh Toan Nguyen, Phi Le Nguyen, Hongzhi Yin, and Quoc Viet Hung Nguyen. 2024. A Survey of Privacy-Preserving Model Explanations: Privacy Risks, Attacks, and Countermeasures. *arXiv:2404.00673* (2024).

- [10] Abdullah Caglar Oksuz, Anisa Halimi, and Erman Ayday. 2024. AUTOLYCUS: Exploiting Explainable Artificial Intelligence (XAI) for Model Extraction Attacks against Interpretable Models. *Proceedings on Privacy Enhancing Technologies (PoPETs) 2024*, 4 (2024). arXiv:2302.02162.
- [11] Abdullah Caglar Oksuz, Anisa Halimi, and Erman Ayday. 2026. LoMime: Query-Efficient Membership Inference using Model Extraction in Label-Only Settings. *arXiv:2602.18934* (2026).
- [12] Tribhuvanesh Orekondy, Bernt Schiele, and Mario Fritz. 2019. Knockoff Nets: Stealing Functionality of Black-Box Models. In *CVPR*.
- [13] Daan Otto, Jannis Kurtz, Dick den Hertog, and Ilker Birbil. 2026. Linear Model Extraction via Factual and Counterfactual Queries. *arXiv:2602.09748* (2026).
- [14] Soham Pal, Yash Gupta, Aditya Shukla, Aditya Kanade, Shirish K Shevade, and Vinod Ganapathy. 2020. ACTIVETHIEF: Model Extraction Using Active Learning and Unannotated Public Data. In *AAAI*.
- [15] Nicolas Papernot, Patrick McDaniel, Ian Goodfellow, Somesh Jha, Z Berkay Celik, and Ananthram Swami. 2017. Practical Black-Box Attacks Against Machine Learning. In *AsiaCCS*.
- [16] Florian Tramèr, Fan Zhang, Ari Juels, Michael K Reiter, and Thomas Ristenpart. 2016. Stealing Machine Learning Models via Prediction APIs. In *USENIX Security*.
- [17] Yongjie Wang, Hangwei Qian, and Chunyan Miao. 2022. DualCF: Efficient Model Extraction Attack from Counterfactual Explanations. In *ACM Conference on Fairness, Accountability, and Transparency (FAccT)*. doi:10.1145/3531146.3533188 arXiv:2205.06504.
- [18] Anli Yan, Teng Huang, Lishan Ke, Xiaozhang Liu, Qi Chen, and Changyu Dong. 2023. Explanation Leaks: Explanation-guided Model Extraction Attacks. *Information Sciences* 632 (2023), 269–284. doi:10.1016/j.ins.2023.03.020